

Problem 16

Are the events A and $\overline{A + B + C}$ compatible (can they occur simultaneously)?

Solution

Two events are compatible if they can occur simultaneously, i.e., if their intersection is non-empty: $A \cap \overline{A + B + C} \neq \emptyset$.

Step 1: Simplify $\overline{A + B + C}$

Using De Morgan's law:

$$\overline{A + B + C} = \bar{A} \cdot \bar{B} \cdot \bar{C}$$

This represents the event where none of A , B , or C occur.

Step 2: Find the Intersection

We need to check if:

$$A \cap \overline{A + B + C} = A \cap (\bar{A}\bar{B}\bar{C})$$

is non-empty.

Step 3: Compute the Intersection

$$A \cap (\bar{A}\bar{B}\bar{C}) = A \cdot \bar{A} \cdot \bar{B} \cdot \bar{C}$$

Using the complement law: $A \cdot \bar{A} = \emptyset$

$$= \emptyset \cdot \bar{B} \cdot \bar{C} = \emptyset$$

Step 4: Conclusion

Since $A \cap \overline{A + B + C} = \emptyset$, the events are ****incompatible**** (mutually exclusive).

NO, the events are NOT compatible

Explanation

Intuitive reasoning:

- Event A means "A occurs"
- Event $\overline{A + B + C}$ means "none of A , B , or C occur"
- For both to occur simultaneously, we would need A to occur AND A not to occur
- This is a logical contradiction

Alternative Verification

We can also verify this using logical reasoning:

If A occurs, then $A + B + C$ must occur (since $A \subseteq A + B + C$).

Therefore, if A occurs, then $\overline{A + B + C}$ cannot occur.

Conversely, if $\overline{A + B + C}$ occurs, it means $A + B + C$ doesn't occur, which implies A doesn't occur.

Hence, the two events cannot occur simultaneously.

Formal Proof

To prove: $A \cap \overline{A+B+C} = \emptyset$

Proof:

Suppose $\omega \in A \cap \overline{A+B+C}$.

Then:

- $\omega \in A$ (by definition of intersection)
- $\omega \in \overline{A+B+C} = \overline{A}\overline{B}\overline{C}$ (by definition of intersection)

From the second condition: $\omega \in \overline{A}$, which means $\omega \notin A$.

But this contradicts the first condition that $\omega \in A$.

Therefore, no such ω exists, and $A \cap \overline{A+B+C} = \emptyset$.

Probabilistic Interpretation

If we assign probabilities:

$$P(A \cap \overline{A+B+C}) = P(\emptyset) = 0$$

The probability that both events occur simultaneously is zero, confirming they are incompatible.

Venn Diagram Interpretation

In a Venn diagram:

- A is represented by the circle/region labeled A
- $A+B+C$ is the union of all three regions
- $\overline{A+B+C}$ is everything outside all three circles
- The intersection of A (inside the A circle) and $\overline{A+B+C}$ (outside all circles) is empty

Related Concept

Events X and Y are:

- **Compatible (can occur together)** if $X \cap Y \neq \emptyset$
- **Incompatible (mutually exclusive)** if $X \cap Y = \emptyset$

In our case, A and $\overline{A+B+C}$ are incompatible.

General Result

More generally, for any event A and any set S containing A :

$$A \cap \overline{S} = \emptyset \quad \text{if } A \subseteq S$$

Since $A \subseteq A+B+C$, we have:

$$A \cap \overline{A+B+C} = \emptyset$$

Final Answer

NO - The events A and $\overline{A+B+C}$ are INCOMPATIBLE (mutually exclusive)

Reason: $A \cap \overline{A+B+C} = A \cap \overline{A}\overline{B}\overline{C} = \emptyset$

The event A requires A to occur, while $\overline{A+B+C}$ requires A NOT to occur. These conditions are contradictory, so the events cannot occur simultaneously.

Key insight: If A occurs, then automatically $A+B+C$ occurs, which means $\overline{A+B+C}$ cannot occur.